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Cut it into blocks: how Hoover Dam was cooled

The free guide to the reel on Hoover Dam: why setting concrete gets hot, why a dam poured in one piece would have needed about 125 years to cool, how blocks and cold-water pipes solved it, and how to see the effect at home.



Why concrete gets hot

Cement reacts with water as it hardens (hydration), and that reaction gives off heat. In a footpath the heat escapes in hours. In a mass of concrete hundreds of metres thick, the heat is trapped in the middle.

As the inside slowly cools, it shrinks while the outside is already cold and stiff. The difference in shrinkage pulls the concrete apart and opens cracks, the worst thing that can happen to a dam.

The Hoover Dam numbers

The US Bureau of Reclamation calculated that, poured as one continuous mass, the dam (completed in 1935, about 2.5 million cubic metres of concrete) would have taken about 125 years to cool to the temperature of its surroundings.

What the engineers did

The dam was built as columns of interlocking blocks, each poured in lifts of about 1.5 m, so that every block could cool on its own.

More than 580 miles (about 930 km) of 1-inch steel pipe were laid in the concrete. Cold river water was pumped through them first, then water chilled by a refrigeration plant built on site. When a block had cooled and stopped shrinking, its pipes and the joints between blocks were filled with grout under pressure, turning the blocks back into one monolith. Cooling was complete in 1935.

Built examples of controlling concrete heat

Hoover Dam, USA (1933-1935)	Blocks plus embedded cooling pipes carrying river water and refrigerated water.
Grand Coulee Dam, USA (1942)	Used the same block-and-pipe method on an even larger volume of concrete.
Modern mass pours (foundations, dams)	Cooling pipes are still used, together with low-heat cement, ice or chilled water in the mix, and temperature sensors inside the pour.

Try it yourself: why size matters for cooling

1. Fill a large jar and a small glass with hot tap water at the same temperature.
2. Put a kitchen thermometer in each and note the temperature every 5 minutes for half an hour.
3. The small glass cools much faster: it has more surface for its volume.
4. Now split the large jar's water into four small glasses and repeat: together they cool as fast as the small one.
5. Finally, put a thin metal tube (or a straw with ice water) through a full jar: the water around it cools much faster. That is the Hoover Dam idea: smaller blocks, and a cold pipe through each one.

The rule to remember

When a process is too big to control in one go, cut it into parts you can watch and correct one by one.

Default answer	Do it all at once and hope it behaves.
Form-finder answer	Split it into blocks, give each one its own control, then join them.
Check	Ask: what goes wrong because of the size alone?

Go further

Bureau of Reclamation: Hoover Dam concrete
https://www.usbr.gov/lc/hooverdam/History/essays/concrete.html

Hoover Dam (Wikipedia)
https://en.wikipedia.org/wiki/Hoover_Dam

NRMCA: Hoover Dam, a historic concrete piece
https://www.nrmca.org/america250/hoover-dam/

Credits
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Facts: US Bureau of Reclamation, NRMCA, Wikipedia.

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